can function only if there is a good supply of genetic variation to work upon. If you try to breed, say, rabbits for ever longer ears, to begin with you'll succeed. The average rabbit in a wild population will have medium-sized ears (by rabbit standards; by our standards, of course, it will have very long ones). A few rabbits will have shorter than average ears and a few longer than average. By breeding only from those with the longest ears you'll succeed in increasing the average in later generations. For a while. But if you *continue* to breed from those with the longest ears there will come a time when the necessary variation is no longer available. They'll all have the 'longest' ears, and evolution will grind to a halt. In normal evolution this sort of thing is not a problem, because most environments don't carry on exerting consistent and unswerving pressure in one direction. The 'best' length for any particular bit of an animal will normally not be 'a bit longer than the present average, whatever the present average may be'. The best length is more likely to be a fixed quantity, say three inches. But sexual selection really can have the embarrassing property of chasing an ever-driving 'optimum'. Female fashion really could desire ever longer male ears, no matter how long the ears of the current population might already be. So variation really could seriously run out. And yet sexual selection does seem to have worked; we do see absurdly exaggerated male ornaments. We seem to have a paradox, which we may call the paradox of the vanishing variation.

Lande's solution to the paradox is mutation. There will always be enough mutation, he thinks, to fuel sustained selection. The reason people had doubted this before was that they thought in terms of one gene at a time: mutation rates at any one genetic locus are too low to resolve the paradox of vanishing variation. Lande reminded us that 'tails' and other things that sexual selection works on are influenced by an indefinitely large number of different genes —'polygenes'—whose small effects all add up. Moreover, as evolution goes on, it will be a shifting set of polygenes that are relevant: new genes will be recruited into the set that influence variation in 'tail length', and old ones lost. Mutation can affect any of this large and shifting set of genes, so the paradox of vanishing variation itself vanishes.

W. D. Hamilton's answer to the paradox is different. He answers it in the same way he answers most questions nowadays: 'Parasites'. Think back to the rabbit ears. The best length for rabbit ears presumably depends on various acoustic factors. There is no particular reason to expect these factors to change in a consistent and sustained direction as generations go by. The best length for rabbit ears may not be absolutely constant, but still selection is unlikely to push so far in any particular direction that it strays outside the range of variation easily thrown up by the present gene pool. Hence no paradox of vanishing variation.

But now look at the kind of violently fluctuating environment provided by parasites. In a world full of parasites there is strong selection in favour of ability to resist them. Natural selection will favour whichever individual rabbits are least vulnerable to the parasites that happen to be around. The crucial point is that these will not always be the same parasites. Plagues come and go. Today it may be myxomatosis, next year the rabbit equivalent of the black death, the year after that rabbit AIDS, and so on. Then, after say a ten-year cycle it may be back to myxomatosis, and so on. Or the myxomatosis virus itself may evolve to counter whatever counteradaptations the rabbits come up with. Hamilton pictures cycles of counteradaptation and counter-counteradaptation endlessly rolling through time and forever perversely updating the definition of the 'best' rabbit.

The upshot of all this is that there is something importantly different about adaptations for disease-resistance as compared with adaptations to the physical environment. Whereas there may be a pretty fixed 'best' length for a rabbit's legs to be, there is no fixed 'best' rabbit as far as disease-resistance is concerned. As the currently most dangerous disease changes, so the currently 'best' rabbit changes. Are parasites the only selective forces that work this way? What about predators and prey, for instance? Hamilton agrees that they are basically like parasites. But they don't evolve so fast as many parasites. And parasites are more likely than predators or prey to evolve detailed gene-for-gene counteradaptations.